CIVIC PLAN

LAND MANAGEMENT

SMART DIGITAL LAND MANAGEMENT AND PLANNING APPROVAL SYSTEM

Supervisor - Ms. Inoshi Jayaweeera

Applied Project - CSG3101.2

Assignment 1: Project Proposal

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PROPOSED BY SRI_GROUP_43

Mora Mudalige Thenuk Sandul - 10674268

R.A.D. Akash Dhananjaya Randeniya - 10659913

Niman Nethmika Rathnayake - 10660786

W. Nadeeja Ayeshan Weerasekara - 10685137

K.D. Prashan Kalhara Wijesinghe - 10715485

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1. Project goal

The goal of this project is to develop a secure, centralized Digital Land Management and Planning Approval System that enables efficient user authentication, property record handling, automated and manual planning approvals, GIS-based land visualization, and comprehensive document and workflow management for citizens and administrators.

2. Background

In Sri Lanka, individuals who wish to construct a building must obtain planning approval from the local Municipal Council or the relevant Urban Development Authority (UDA). For this approval to be granted, building plans must comply with UDA regulations, and the ownership of the land must be properly registered with the Land Registry (*8 Things to Follow to Get Your House Plan Approved by the Municipal Council*, n.d.). However, the existing approval process is highly manual, paper-based, and often inefficient, leading to significant delays for citizens and administrative officers.

A major issue in the current system is that citizens do not receive an initial approval or rejection based on the documents they submit. If required documents are missing or incomplete, it typically takes one to two weeks for the local authority to notify the applicant. This lack of immediate feedback causes unnecessary delays, often extending the entire approval timeline by two to three additional weeks (*The Urban Development Authority*, n.d.). The process becomes repetitive and time-consuming for both citizens and government officers, contributing to administrative congestion and frustration.

Additionally, citizens who need to determine the current value of their property must rely on professional values. Obtaining an official valuation report is expensive, often costing around 1.5% of the property value, and takes several weeks to complete Procons (n.d.). There is currently no accessible or cost-free method for landowners to instantly understand the approximate or predicted land value without going through lengthy professional procedures.

These challenges highlight a significant gap in Sri Lanka's land and planning approval administrative framework. The absence of a streamlined, responsive, and transparent digital mechanism results in prolonged approval cycles, increased workload for officers, added financial

burden for citizens, and limited access to essential property information. A modernized system is required to address these inefficiencies and improve the overall experience of citizens seeking planning approval.

3. Project Scope

The Smart Digital Land Management and Planning Approval System aims to digitize traditional land management and planning approval processes by replacing manual, paper-based workflows with a secure digital platform. The system will allow citizens to register, access personalized dashboards, manage property records, and submit planning approval documents, while administrators can review, track, approve, or reject submissions efficiently. By centralizing land records and approval procedures, the platform improves transparency, reduces administrative delays, and simplifies land-related services for both citizens and government authorities.

The system will also include additional features such as GIS-based land parcel visualization, property record management, document validation, real-time reporting, and automated notifications. Furthermore, it will support land analysis functions, including current land valuation and future land price prediction based on historical and geographical data. With integrated security mechanisms to protect user information, the platform aims to reduce paperwork, improve administrative workflows, and provide a modern and reliable solution for managing land records and planning approvals.

3.1. In Scope

1. User registration, login, password reset, Admin login, and account management
2. User profile updates
3. Suspicious login monitoring and notifications
4. Document submission for planning approval
5. Automatic rule-based approval/rejection, Manual admin approval/rejection, and override
6. Requesting additional documents
7. Viewing property records and transaction history

8. Ownership history report generation
9. Admin search (Land ID, NIC, Case ID, Owner Name)
10. GIS map to show land location
11. Basic current & future land value estimation
12. Daily/weekly/monthly case statistics, Daily report on submissions/approvals/rejections
13. Pending tasks and deadline alerts
14. Access to support documents, rules, and gazettes

3.2. Out of Scope

1. Mobile app development (Android/iOS)
2. Integration with real government databases (Land Registry, UDA, MC)
3. Advanced AI chatbot with NLP
4. Online payment gateway
5. Legal dispute or conflict resolution workflows
6. Physical land surveying or field inspection features
7. Full multi-language support (Sinhala/Tamil)

3.3. Requirements

3.3.1. Functional Requirements

Please refer to the Functional Requirements

3.3.2. Non-Functional Requirements

Please refer to the Non-Functional Requirements

4. Table of Deliverables

Deliverable	Description	Format Output	Due Date
Project Proposal	A document outlining the background project, objectives, scope, requirements, methodology, and planned schedule for the system.	Word and PDF Document	15 March 2026
System Design and Modelling Document	A document containing all system design diagrams, including UML models, ERD, wireframes, and architectural diagrams that describe the system's structure and behavior.	PDF Document	21 March 2026
Source Code	Complete and documented source code with version control.	Zip file containing all source code	03 May 2026
User Guide	Guide explaining how to use the system: login, submit documents, track approvals, etc.	PDF Document	06 May 2025
Administrator Guide	Detailed instructions for admin operations, approval workflow, user management, and handling special cases	PDF Document	06 May 2026
Developer Manual	A manual describing system architecture, coding structure, and maintenance guidelines.	PDF Document	06 May 2026
System Test Results	Documentation of all test cases, results, and identified bugs.	PDF Document	06 May 2026
Survey Report	A report summarizing user or stakeholder feedback collected through surveys, including analysis of user experience, satisfaction, and system usability.	PDF Document	06 May 2026
Project Presentation	A slide presentation highlighting the project's overview, requirements, design, and results.	Microsoft PowerPoint Document	01 May 2026
Final Report	A report detailing how the system meets the requirements outlined in the proposal.	PDF Document	15 May 2026

5. Schedule

Week No.	Main Task	Start Date	End Date	Deliverable

1	Project initiation, requirement gathering	Week 1 Day 1	Week 1 Day 7	Draft requirements document
2	System analysis & use case modelling	Week 2 Day 1	Week 2 Day 7	Use case diagrams + Analysis document.
3	ERD, class diagrams, activity diagrams	Week 3 Day 1	Week 3 Day 7	ERD + UML diagrams
4	UI/UX wireframes and architecture finalization	Week 4 Day 1	Week 4 Day 7	Wireframes + Architecture diagrams
5	Backend development (Authentication & User Management)	Week 5 Day 1	Week 5 Day 7	Authentication API module
6	Backend development (Document submission + validation engine)	Week 6 Day 1	Week 6 Day 7	Document handling backend
7	Frontend development (User Dashboard + Admin Dashboard)	Week 7 Day 1	Week 7 Day 7	Frontend dashboard screens
8	GIS integration & land valuation module	Week 8 Day 1	Week 8 Day 7	GIS module + valuation module
9	Security implementation & system integration	Week 9 Day 1	Week 9 Day 7	Integrated system build
10	Functional & non-functional testing	Week 10 Day 1	Week 10 Day 7	Test cases + bug report
11	Documentation & refinements	Week 11 Day 1	Week 11 Day 7	User guide, Admin guide, Developer manual
12	Deployment & final presentation	Week 12 Day 1	Week 12 Day 7	Final system + Presentation + Final Report

Please refer to the Gantt Chart: [Final Smart Digital Land Management and Planning Approval System Gantt Chart.mpp]

6. Team Capability Alignment

Team Member	Role in Project	Course Enrolled	Relevant Course Learning Outcomes	Tasks in the Project
Mora Mudalige Thenuk Sandul	Data Analyst Software Developer 1 Project Manager 2	Software Engineering	Apply software design principles and system architecture	Requirements gathering, system modelling, architecture design, land valuation model development, task allocation, project scheduling, stakeholder communication
R.A.D. Akash Dhananjaya Randeniya	Software Developer 2 UI/UX Designer Project Manager 1	Software Engineering	Develop secure server-side applications and database integration	Front-end and UI/UX design, dashboard development, implementing user authentication module, API development, assisting in project planning and progress tracking
Niman Nethmika	Software Developer 3 Quality Assurance	Software Engineering	Design and manage relational databases, ensuring data integrity	Database de sign (ERD, schema creation), SQL query development, document validation engine integration, test case creation, system testing, defect reporting
Nadeeja Ayeshan	Security Architect IAM Specialist Security Auditor	Cyber Security	Identify vulnerabilities and implement security controls	Designing security architecture, implementing RBAC and MFA, monitoring login anomalies, conducting vulnerability assessments, and ensuring compliance with security standards
–Prashan Kalhara	Application Security Engineer Penetration Tester Security Consultant	Cyber Security	Analyze data and apply analytical tools for decision support	Performing penetration testing, secure coding review, evaluating system logs, supporting the development of automated suspicious activity detection, and advising on threat mitigation

7. Tools and Technical Requirements

7.1. Frontend development

Languages: HTML, CSS, and JavaScript

- These technologies will be used to create a responsive and user-friendly interface. HTML will structure the web pages, CSS will be used for styling, and JavaScript will add interactivity and handle client-side logic.

7.2. Backend development

Language: Python

- Python will be utilized for server-side scripting, ensuring scalability and easy integration with databases and APIs. It will handle all backend processes such as user authentication, document validation, and data processing.

7.3. Database

Language: MySQL

- MySQL will be used to store and manage land records, planning approvals, and user data. It will provide a secure and structured database solution for handling the large amount of data expected by the system.

7.4. GIS mapping integration

Technologies: Google Maps API / OpenStreetMap

GIS (Geographic Information System) technology will be integrated to visualize land parcels and geographic data.

- Google Maps API will be used to display maps within the system interface.

7.5. Development and collaboration

Software: Visual Studio Code

- IDE for development

Tool: GitHub

- GitHub will be used for version control, enabling collaboration between the team members, tracking changes, and maintaining a backup of the project source code

7.6. Document processing tools

- Python PDF libraries (ReportLab / PyPDF2) for generating approval or rejection letters.

7.7. Additional tools

- Microsoft Word / Google Docs for preparing project documentation and reports.

- Draw.io/ Lucid chart for creating system diagrams such as UML, ERD, and architecture diagrams.

8. Future Enhancements

These features are not part of the current project but may be added in future releases. They aim to extend system capabilities, improve user experience, and support larger-scale government integration.

- Mobile Application (Android/iOS) offers push notifications and real-time status updates.
- Full Integration with Government Databases for automatic verification of land ownership and deed details.
- Advanced GIS Mapping with zoning layers, road networks, flood zones, and shapefile datasets.
- Payment Gateway Integration for planning fees, survey fees, and certification charges.
- Role-Based Workflow Automation for planning officers, supervisors, surveyors, and legal reviewers.
- Multi-Language Support (Sinhala & Tamil) for improved accessibility across all regions.
- Analytics Dashboards display KPIs, processing times, land case density, and departmental performance.

9. References

8 things to follow to get your house plan approved by the Municipal Council. (n.d.). LankaPropertyWeb.com News & Blog. https://www.lankapropertyweb.com/property-news/8-things-to-follow-to-get-your-house-plan-approved-by-the-municipal-council/

Procons. (n.d.). *Valuation fees circular*. https://valuationdept.gov.lk/index.php?option=com_content&view=article&id=52&Itemid=215&lang=en

The Urban Development Authority. (n.d.). https://www.uda.gov.lk/approval-process.html

10. Appendix

Appendix 1 - Functional Requirements

Functional Requirnments.xlsx

Functional Requirements			
Main Category	NO:	Priority	Background
Account Management	FR01	Must	Users must be able to register an account using their NIC number, full name, property address, date of birth, city, email, and password.
	FR02	Must	Registered users must be able to log into their accounts using NIC number and password.
	FR03	Must	Administrators must be able to log into their portal using full name as the username and employee ID as the password.
	FR04	Must	The system must provide a "Forgot Password" feature that allows users to reset their password through a password reset link sent to their registered email address.
	FR05	Must	Users must be able to redirect their personalized dashboard after successful login.
User Profile Management	FR06	Must	Users must be able to update their personal details, including name, address, contact information, and password.
Security and Suspicious Activity Monitoring	FR07	Should	The system should monitor login attempts to detect suspicious activities, such as multiple failed login attempts.
	FR08	Should	The system should alert administrators when suspicious login activity is detected.
	FR09	Could	The system could send email or SMS notifications when a successful login occurs from an unrecognized device or location.
	FR10	Must	Administrators must be able to suspend or deactivate user accounts if suspicious activity is detected.
	FR11	Must	Administrators must be able to modify or delete standard user accounts.
Property Record Management	FR12	Should	Users should be able to view their property records and transaction history of their properties.
	FR13	Must	The system must display pending requests and alerts related to user property records.
	FR14	Must	Administrators must be able to search property records using Land ID, owner name, NIC number and Case ID.
	FR15	Must	The system must generate reports of previous landowners, including their name, address, phone number, and deed transfer date.
Document Submission and Management	FR16	Must	Users must be able to submit documents required for planning approval.
	FR17	Must	Administrators must be able to review submitted planning approval documents.

	FR18	Must	Administrators must be able to preview uploaded documents within the system without downloading them.
	FR19	Must	Administrators must be able to request additional documents if submitted documents are incomplete.
	FR20	Must	The system must notify users when additional documents are requested, submitted documents are approved or rejected .
Duplicate and Validation Checks	FR21	Should	The system should detect duplicate land plan submissions for the same land parcel and alert administrators.
	FR22	Must	The system must automatically validate submitted documents using predefined planning approval rules.
	FR23	Must	The system must automatically approve documents when all validation rules are satisfied or reject documents when validation rules are violated.
	FR24	Must	Administrators must be able to manually approve or reject submitted documents .
Document Repository and Letters	FR25	Must	The system must generate an approval or rejection letter when a document is approved or rejected.
	FR26	Could	The system could store all approved and rejected documents in the system repository.
	FR27	Could	The system could store all generated approval and rejection letters sent to landowners.
	FR28	Could	The system could allow users to request a certified copy of an approval document using the document ID.
Reporting and Tracking	FR29	Must	The system must display the number of uploaded cases daily, weekly, and monthly.
	FR30	Must	The system must generate a daily report showing the total number of cases submitted, approved and rejected cases.
	FR31	Must	The system must display pending administrative tasks requiring attention.
	FR32	Must	The system must display property records that require verification or updates.
	FR33	Must	The system must track planning approval deadlines and notify administrators if a case remains pending for too long.
GIS Mapping and Land Valuation	FR34	Should	The system should display land parcel locations using GIS mapping.
	FR35	Should	The system should generate the current land value for a property.
	FR36	Should	The system should predict future land values based on available data.
Supporting Documents and Policies	FR37	Must	The system must provide the supporting documents required for planning approval.
	FR38	Should	The system should provide gazettes, rules, and policies related to planning approval.
User Assistance	FR39	Could	The system could include a chatbot to answer frequently asked questions related to land management and planning approval.

Appendix 2 - Non-Functional Requirements

Non-Functional Requirements.xlsx

Non-Functional Requirements			
Main Category	NO:	Priority	Background
Security Requirements	NFR01	Must	All communication between users and the system must be encrypted using HTTPS/TLS.
	NFR02	Must	User passwords must be securely stored using strong hashing algorithms (e.g., encrypt or SHA-256).
	NFR03	Must	The system must implement role-based access control (RBAC) to restrict access for citizens and administrators.
	NFR04	Must	The system must prevent common web attacks such as SQL Injection, Cross-Site Scripting (XSS), and Cross-Site Request Forgery (CSRF).
	NFR05	Should	Multi-factor authentication should be enabled for administrators.
Performance Requirements	NFR06	Could	The system could be able to return property search results within 3 seconds.
	NFR07	Should	GIS map data and land parcel boundaries should load within 2 seconds.
	NFR08	Should	The system should support at least 1,000 concurrent users without significant performance degradation.
	NFR09	Must	Document uploads and downloads must complete within 5 seconds for files under 10MB.
Data Integrity Requirements	NFR10	Must	The database must enforce ACID transaction properties to prevent incomplete updates.
	NFR11	Must	All property transactions must be validated before being stored in the database.
Audit and Logging Requirements	NFR12	Should	The system should be able to record all important user activities such as login, document submission, approval, and rejection.
	NFR13	Could	Audit logs could include user ID, timestamp, action performed, and affected records.
	NFR14	Must	Logs must be securely stored and accessible only to authorized administrators.
Usability Requirements	NFR15	Must	The platform must support responsive design for desktops, tablets, and mobile devices.
	NFR16	Must	The system must support the English language.
Reliability Requirements.	NFR17	Must	The system must detect errors and notify administrators.

Appendix 3 - Team Contract

CSG3101 Group Contract.docx.pdf

Group Contract for CSG3101 Applied Project

Group norms are the rules that define acceptable behavior amongst members of a group. Norms include levels of performance valued by the group, expectations of group members, beliefs and values in relation to study, relationships between group members and teamwork within the group. Writing down these agreed norms in the form of a Group Contract provides a means of clarifying and enforcing these norms when necessary. This generally leads to higher levels of commitment to group goals and better group performance. It can also reduce the chances of the ***free-rider*** emerging within the group which can be distressing for all group members, particularly responsible group members who do their fair share of the work.

A ***Group Contract*** is a requirement of this unit and must be submitted in your first Project Management Report complete with all signatures.
A sample of the Group Contract is included below which must be modified according to your group's needs.

Expulsion of a Group Member: In cases where one or more members act contrary to the objectives of the whole group a method of dispute resolution (and in some cases expulsion) must be included in the contract. In the event that the majority of a group wishes to expel a group member, they must show clearly the number of opportunities that the member in question has received to remedy a situation. This must be verifiable and auditable (note verbal acknowledgment is insufficient). There must be a minimum of three occasions where a group member has been notified of his/her inappropriate performance, and the issues should be clearly documented in previous weekly progress reports showing both a member's poor performance AND the means by which this was professionally communicated within the group.

Sample contract:

Group Contract

1. The subject matter of this contract

This contract is entered into by the students named below for the purpose of ensuring that each individual group member fulfills his/her obligations for completing the group assignment for the unit CSG3101 Applied Project.

2. The consideration

1. All group members will be punctual at meetings.
2. All group members should attend meetings unless by prior agreement with the group.
3. All group members will stay at the meeting until it is agreed that the meeting is adjourned.
4. All group members will agree to a specific day/time for each weekly meeting and an agreed procedure for informing all other members prior to any missed meetings.
5. All group members will come to the meetings prepared by completing the agreed tasks on time.
6. The group will actively seek the contributions and opinions of each member at meetings and during group discussions.
7. Each group member will take turns at both listening and talking.
8. Dominating the group's discussion and decision making is not acceptable.
9. Group members will take turns in writing down minutes of the meeting.
10. The group member taking minutes will record allocated tasks to be completed by group members by name and agreed deadlines for task completion.
11. The group members must decide and declare which method of communication is to be the preferred and agreed method (i.e., MS Teams, ECU email, face-to-face.
12. The group members must decide on an auditable and verifiable method of detailing completed tasks and or individual elements. (i.e., Google groups, tasks management, written master file)
13. Work allocation will be according to an agreed procedure and is documented below *(insert method of breaking down the assignment and allocation of work to group members).*
14. Where disputes arise regarding the work tasks or agreed behaviours in which a group member is not performing according to the terms of this agreement, the following process will be entered into to resolve the dispute: *(insert the dispute resolution procedure for your group*). Disputes must be resolved within the group and documentation must be retained that relates to attempts to resolve a dispute or to encourage a group member to make his or her contribution to the assignment).
15. Ejection of a non-performing or disruptive group member. The Contract MUST include a specific method for dealing with group members where the performance of the group is affected.

3. Names and signatures of the parties to this Group Contract

Student signature	Student name (printed)	Student Number	Date
	Mora Mudalige Thenuk Sandul	10674268	10/03/2026
	R.A.D. Akash Dhananjaya Randeniya	10659913	10/03/2026
	Niman Nethmika Rathnayake	10660786	10/03/2026
	Nadeeja Ayeshan	10685137	10/03/2026
	Prashan Kalhara	10715485	10/03/2026

Date of Fully Signed copy given to Supervisor and Unit Coordinator

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